

GROUP A — Short Answer Questions (5 Marks Each)

1. What is voltage? Define potential difference and write its formula.
2. Define electric current and write its formula. What are the possible charge carriers responsible for current?
3. What is resistance? State and explain Ohm's Law.
4. What is a resistor? Explain any two uses of a resistor in an electronic circuit.
5. What is drift velocity? Explain with the help of a diagram how it differs from the random motion of electrons.
6. Define mobility of electrons. Write its formula and unit.
7. Explain hydroelectric power generation with the help of a labeled diagram.
8. What is solar energy? Explain the working of a photovoltaic cell.
9. Differentiate between conventional flow of current and modern flow of current.
10. Write short notes on: (a) Resistivity (b) Conductance.

GROUP B — Long Answer Questions (8 Marks Each)

1. What is nuclear energy? Explain Nuclear Fusion and Nuclear Fission with their respective reaction equations and examples.
2. Derive the formula for drift velocity of electrons in a conductor starting from the force experienced due to an electric field. Also solve: Calculate the resistance of a copper wire of length 5 m and cross-sectional area 0.01 m^2 ($\rho_{\text{cu}} = 1.72 \times 10^{-8} \Omega \cdot \text{m}$).
3. Explain the factors affecting the resistance of a conductor. Derive the resistivity formula $R = \rho l/A$.
4. Classify materials on the basis of resistance into insulators, conductors, and semi-conductors. Explain each with energy band diagrams and examples. What is doping?
5. Explain the sources of electricity. Describe wind energy and thermal energy (Seebeck Effect) as sources of electricity generation.

*** Best of Luck! ***